FPT ACADEMY INTERNATIONAL

FPT – APTECH COMPUTER EDUCATION

MediFlow

A Microservices – Based Distributed Hospital Management System

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Semester: *04*

Batch No: *T2.2410.E0*

Group No: *04*

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***Month**: 10 **Year**: 2026*

This is to certify that

Mr. PHAM DANG VINH

Mr. LE QUANG HUY

Mr. TRAN HOANG ANH

Mr. THAN THE LOC

Have successfully Designed & Developed

MEDIFLOW

A MICROSERVICES – BASED DISTRIBUTED HOSPITAL MANAGEMENT SYSTEM

Submitted by:

Mr. HOANG DUC QUANG

Date Of Issue:

Authorized Signature:

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Acknowledge

As we approach the completion of our **MediFlow – Microservices-Based Distributed Hospital Management System** project, we are filled with gratitude, pride, and a sense of accomplishment. This project has been a valuable opportunity to apply our academic knowledge, strengthen our technical skills, and gain practical experience in developing complex software systems.

Heartfelt Gratitude to Our Mentors

First and foremost, we would like to express our sincere gratitude to the **Project Team at FPT Aptech** for providing us with the knowledge, resources, and opportunity to undertake this meaningful project.

We are especially thankful to our lecturer, **Mr. LE THANH NHAN**, for his dedicated guidance, patience, and valuable feedback throughout the development process. His mentorship has helped us overcome technical challenges, improve our problem-solving abilities, and develop a more professional approach to software engineering.

Appreciation for Our Team and Families

We extend our heartfelt appreciation to every member of **Team 04** for their dedication, teamwork, and perseverance. Through close collaboration, we have explored and developed a distributed healthcare platform encompassing ten business domains: **Organization, Patient, Clinical, Laboratory, Pharmacy, Billing, Notification, Reporting, Inpatient, and Surgery**.

We are equally grateful to our families, friends, and classmates for their continuous encouragement and understanding throughout this journey.

A Meaningful Learning Experience

The **MediFlow Project** has provided us with valuable experience in microservices architecture, distributed systems, database design, event-driven communication, and cross-platform application development. By working with technologies such as **Spring Boot, Spring Cloud, RabbitMQ, PostgreSQL, Next.js, and Flutter**, we have gained a deeper understanding of modern software engineering practices and the challenges of integrating independent healthcare services.

Beyond technical knowledge, this project has strengthened our communication, adaptability, responsibility, and collaborative problem-solving skills.

Looking Forward

As we conclude this important milestone, we believe the knowledge and experience gained from MediFlow will serve as a strong foundation for our future careers in information technology.

Once again, we sincerely thank everyone who has supported and contributed to our journey.

With heartfelt gratitude,

TEAM 04

MediFlow – Microservices-Based Distributed Hospital Management System

Synopsis

MediFlow – Microservices-Based Distributed Hospital Management System is a healthcare management project designed to address the fragmentation of patient information, hospital operations, and medical workflows across different departments. The system aims to improve coordination, streamline administrative processes, and support efficient healthcare service delivery through a distributed software architecture.

Built on a **microservices-based and event-driven architecture**, MediFlow encompasses ten business domains: **Organization, Patient, Clinical, Laboratory, Pharmacy, Billing, Notification, Reporting, Inpatient, and Surgery**. These services are designed to manage distinct healthcare responsibilities while enabling coordinated workflows, including patient registration, appointment scheduling, clinical examinations, laboratory testing, prescription management, payment processing, medication dispensing, inpatient care, and surgical procedures.

The project utilizes **Java Spring Boot, Spring Cloud Gateway, Netflix Eureka, RabbitMQ, and PostgreSQL** for backend development, service discovery, asynchronous communication, and data management. **Next.js and Flutter** are used to develop web and mobile applications, providing role-based interfaces for healthcare professionals and administrative personnel.

MediFlow follows essential software engineering principles, including **Domain-Driven Design, Clean Architecture, database-per-service ownership, RESTful API communication, and event-driven integration**. These approaches promote modularity, maintainability, scalability, and separation of responsibilities across independent services.

As an academic project under active development, MediFlow demonstrates the practical application of distributed systems and modern software engineering technologies in the healthcare domain. Beyond its functional scope, the project provides valuable experience in system architecture, collaborative development, integration testing, and solving complex real-world business problems.

Keywords: Hospital Management System, Healthcare Information System, Microservices, Distributed Systems, Event-Driven Architecture, Spring Boot, RabbitMQ, Next.js, Flutter.

Review 01

Problem Definition

1. Introduction and Context

MediFlow coordinates departments throughout an outpatient journey: reception, examination, laboratory tests, prescribing, payment, and dispensing. Records share a patient identity while each department remains responsible for its own data and business decisions. Users access the web application through the API Gateway.

The project focuses on an outpatient workflow that can be verified end to end, including message redelivery and temporary service outages

2. Problem Identification

- Disconnected patient information, appointments, and clinical records cause repeated entry and make progress difficult to track.
- Doctors, the laboratory, and the pharmacy lack consistent handoff status, increasing the risk of missed orders or results.
- Payment status is not clearly tied to each visit, test, or prescription, making service authorization difficult to control.
- Poor coordination of stock, reservations, and dispensing can cause duplicate dispensing or insufficient stock.
- Successful payment followed by dispensing failure requires traceable compensation rather than a lost transaction.
- Reports and notifications lag behind operations; users struggle to distinguish missing data from a measured zero.
- Medical and financial information requires access restrictions based on role, work scope, and record ownership.

3. Project Objectives

1. Link identity, appointments, visits, tests, prescriptions, and payments through explicit references.
2. Enforce the relevant business and financial conditions before examination, test processing, and dispensing.
3. Prevent duplicate financial or stock effects when requests or events are processed again.
4. Update reports and notifications from events while preserving fault tracing and recovery.
5. Define acceptance criteria for successful workflows, access denial, redelivery, and partial failure.

4. Project Scope

Service	Responsibility
Core	Accounts, staff, departments, reference room data, and patient identity
Clinical	Appointments, check-in, records, diagnosis, orders, and outpatient completion
Lab	Accept test requests, track processing, and return results
Pharmacy	Drug catalogue, stock, prescriptions, reservations, and dispensing
Billing	Charges, invoices/payment requests, payments, service clearance, and compensation/refunds
Support	Reporting read models and in-app notifications

Gateway and Eureka are infrastructure; common is a shared library. Core and Support co-host modules while retaining each context's data and business boundaries. The design retains eight active context stores; six services do not imply six databases.

5. Users and Stakeholders

The eight human roles are ADMIN, DOCTOR, NURSE, PHARMACIST, CASHIER, LAB_TECH, MANAGER, and PATIENT. SYSTEM is a service identity for automated processing, not a human login account. Functions for each role are listed in B1.

Stakeholders include patients, reception staff, doctors, the laboratory, the pharmacy, cashiers, management, and the development team. Departments agree on requirements and verify outcomes within their authorized responsibilities.

6. Proposed Solutions

The Next.js/TypeScript/Tailwind web application accesses the backend through Gateway. The backend uses Java 21, Spring Boot, Maven, PostgreSQL, RabbitMQ, and Eureka. Each context owns its data; contexts do not query each other's databases. REST supplies information needed immediately; events support handoffs and asynchronous updates. Outbox delivery, duplicate protection, bounded retries, and dead-letter queues support recovery.

The main workflow is identity → check-in → examination payment → examination → tests if needed → prescription if needed → medication payment → dispensing → outpatient completion → report/notification updates. When tests are ordered, their fees and authorization are checked before protected laboratory actions.

Customer Requirements Specifications

1. Functional Requirements

The following functions are fucntional requirements. Each action must respect endpoint permissions, clinical authority, and record access boundaries. A role does not grant access to every record. This list neither adds APIs nor expands default permissions.

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	Project Group No: 04	LÊ THANH NHÂN
Signature		
Date		

1.1. ADMIN — Administrator

- Manage accounts, roles, active status, and account locking.
- Manage departments, staff, and reference data required for outpatient care.
- Link accounts to staff or patients using an authoritative identity source.
- Inspect system status, history, and processing failures within permitted administration scope.
- Perform authorized recovery operations with an audit trail, without bypassing financial or clinical conditions.

1.2. DOCTOR — Doctor

- View appointments and patient records within the assigned scope.
- Start an examination only with valid EXAM clearance for the exact visit; emergency exceptions use only the supported, audited mechanism.
- Record examination findings, diagnosis, and outpatient disposition.
- Create explicit laboratory orders and view results returned for the corresponding record.
- Author prescription content; Pharmacy validates and owns prescriptions, prices, stock, and reservations.

- Follow test results and dispensing status, and complete the outpatient record when its conditions are met.

1.3. NURSE — Nurse / Reception Support

- Search, create, and update patient information within granted permissions.
- Assist with appointment registration and departmental schedule tracking.
- Perform check-in, update permitted reception status, and hand patients over to doctors.
- View visit information and processing progress needed for care coordination.
- Check-in does not itself authorize examination or replace financial verification.

1.4. LAB_TECH — Laboratory Technician

- View valid existing test requests and permitted related information.
- Start test processing only with unexpired LAB_TEST clearance matching the exact test.
- Update processing status and cancel when the workflow permits.
- Record a non-empty result with a valid performed date and complete the test.

- Publish results to relevant departments through the Lab workflow.

1.5. PHARMACIST — Pharmacist

- Manage and search the drug catalogue, stock, available quantities, and expiry information.
- Receive doctor-authored prescription content and verify patient identity and outpatient clinical context.
- Create/process permitted prescriptions using server catalogue prices and line-level reservations.
- Check PRESCRIPTION clearance for the exact prescription, reservation validity, and stock before dispensing.
- Confirm dispensing; apply the stock effect once only when all lines satisfy the conditions.
- Cancel or expire prescriptions through the supported workflow, release reservations, and report failures to Billing when financial adjustment is required.
- A payment receipt does not automatically replace dispensing authorization in the target workflow.

1.6. CASHIER — Cashier

- Find charges belonging to the exact visit/patient within permitted service scope.

- Create invoices or payment requests from valid charges in the same visit account.
- Record payments and inspect transaction status and corresponding documents.
- Track financial clearance for the exact examination, test, or prescription; do not manually grant access using a patient-wide paid flag.
- Process compensation/refunds through Billing's workflow and authorization when cancellation or dispensing failure requires it.
- Inspect transaction history for reconciliation; do not rewrite or delete history to force a balance match.

1.7. MANAGER — Manager

- View reports on visits, tests, prescriptions, and dispensed quantities.
- View cash received, refunds, and net cash from available authoritative sources.
- Filter reports using criteria supported by the reporting contract.
- Distinguish missing source coverage from zero, and cash received from earned revenue.
- Track report freshness without modifying source business data from reporting screens.

1.8. PATIENT — Patient

- Sign in and access information associated with their own patient identity.
- View and update permitted personal information.
- Request, view, and track their own appointments through supported workflows.
- View their own examination information, test results, and prescriptions approved for patient access.
- View their own charges, invoices, payment status, and dispensing status through available contracts.
- View in-app notifications addressed to them; do not access another patient's records or notifications.

1.9. SYSTEM — Service Identity

- Transfer valid outpatient events between contexts while preserving identifiers and traceability.
- Automatically create charges/service clearance from valid business sources in the selected profile.
- Handle redelivery, retries, and pending-work recovery without duplicate financial or stock effects.
- Update reporting read models and create in-app notifications for the correct recipients.
- Move messages that remain unprocessable after bounded retries to dead-letter handling for inspection.

- Do not substitute service tokens for human accounts on operations requiring an authenticated human actor.

2. Hardware / Software requirements

2.1. Client requirements

2.1.1. Hardware

Client profile	Minimum	Recommended
Staff desktop/laptop	64-bit dual-core CPU, 4 GB RAM, 1366×768 display, stable network	Modern 4-core CPU, 8 GB RAM, Full HD display, wired or reliable Wi-Fi
Patient browser device	Recent smartphone, tablet, or computer with at least 2 GB available memory	Device receiving current OS/browser security updates, 4 GB or more memory
Development workstation	64-bit 4-core CPU, 16 GB RAM, 30 GB free storage	8-core CPU, 32 GB RAM, SSD with 60 GB or more free storage

Client profile	Minimum	Recommended
Shared reception/lab terminal	Keyboard, pointing device, 1366×768 display, reliable network	Full HD display, barcode-capable peripherals where locally required, UPS-backed power

2.1.2. Software

Item	Requirement
Browser	Current Chrome, Edge, Firefox, or Safari with JavaScript, cookies, and HTTPS enabled
Operating system	A vendor-supported Windows, macOS, Linux, Android, or iOS release
Network	Stable access to the deployed HTTPS endpoint through approved firewall/proxy rules
Document tools	Microsoft Word, LibreOffice Writer, or compatible tool for report conversion

Item	Requirement
Mobile status	Responsive browser access is assumed; a complete native mobile client is not currently delivered

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2.2. Server requirements

2.2.1. Hardware

Environment	CPU	Memory	Storage	Intended use
Developer demonstration	8 logical cores	16 GB RAM	40 GB free SSD	Local containers, selected services, frontend, and tests
Team integration	16 logical cores	32 GB RAM	150 GB SSD	Concurrent services, shared testing, sample data, and monitoring
Initial production baseline	24+ logical cores distributed across hosts	64+ GB RAM distributed	500+ GB resilient SSD plus backups	Starting point only; must be validated by performance and recovery tests

3. Acceptance Criteria

- Verify login and reject locked accounts, wrong roles, wrong token types, and cross-patient access.
- Register a patient and appointment and perform check-in; examination is blocked without clearance and permitted with the exact valid clearance.
- Create a test request; missing, wrong, or expired clearance blocks processing; valid results reach Clinical.
- Create a prescription with server prices and reservations; pay for the correct target; dispense once without negative stock or partial order effects.
- Repeated payment requests/events do not duplicate payments, charges, or stock effects.
- Stop/restart Pharmacy and simulate failure after commit; pending work resumes without lost or duplicate effects.
- Cancellation, expiry, dispensing failure, and late payment preserve terminal states; Billing compensates once in the selected mode and does not claim a completed refund without evidence.
- Reports and notifications update asynchronously for the correct recipients; report replay does not resend notifications.
- Run the six-service profile with infrastructure and preserve accepted data/effects during rollback.

3.1.2. Software

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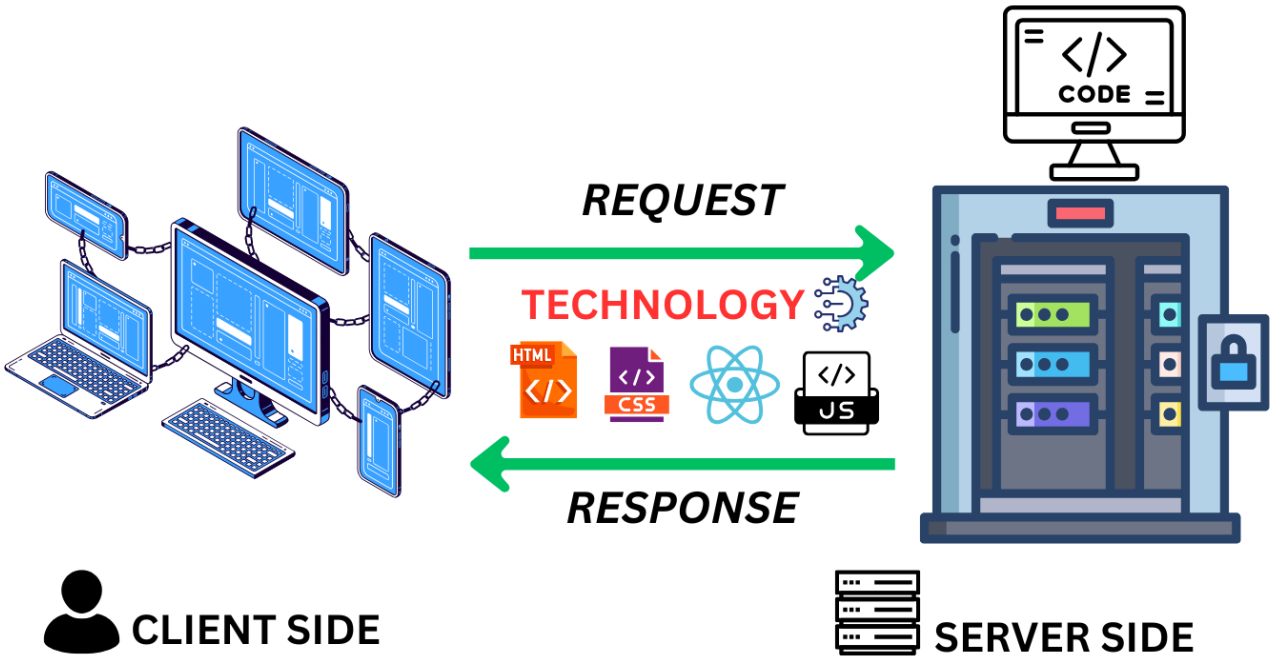
Item	**Source-aligned baseline**
Java	21
Spring Boot	3.3.5
Spring Cloud	2023.0.3
Backend build	Maven multi-module
Frontend	Next.js 16.2.11, React 19.2.4, TypeScript 5, Tailwind CSS 4, pnpm
Database	PostgreSQL 16-compatible, isolated logical database per service
Messaging	RabbitMQ 3-compatible with operational management support
Discovery	Eureka Server
Containers	Docker Engine and Docker Compose compatible with repository definitions
Production platform services	HTTPS certificate management, secret management, centralized logs/metrics/alerts, backups, time synchronization

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Review 02

Architecture Design



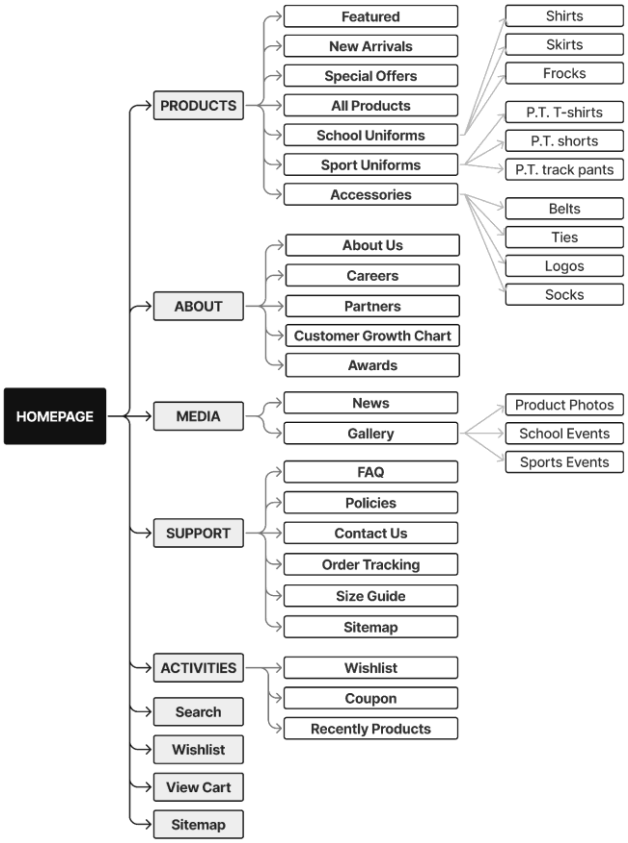
Data Flow Diagram

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Entity Relationship Diagram

Table Design

SiteMap



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Algorithm

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Review 03

Screen shots

1. User Interface Introduction

Before diving into the **Website Structure Overview**, this section provides an overview of the **visual identity, design principles, and tools** used in the **Maverick Dresses** project. These elements ensure consistency, brand recognition, and an optimized user experience across all pages.

2. Brand Identity & Design Concept

2.1. Brand Recognition

Maverick Dresses is a **leading uniform manufacturer**, and the website is designed to reflect its **professional, modern, and trustworthy** identity. The UI/UX approach follows a **minimalist yet elegant** aesthetic, ensuring clarity and ease of navigation for users.

2.2. Color Palette



The color scheme is carefully chosen to balance **brand representation, readability, and visual appeal**. The primary and secondary colors used in the interface include:

- **Primary Colors**:
  - **#f54b49** (Main Accent - Product labels, price highlights)
  - **#eebf8f, #fbece4, #f6eedc** (Supporting Accent - Banners)
  - **#fef2ef** (Top-nav Background & Light Theme Areas)
- **Neutral & Supporting Colors**:
  - **#121212** (Text & Primary Dark UI Elements or Call-to-action button backgrounds)
  - **#454545** (Secondary Text, Menu, Footer)
  - **#7b7b7b** (Disabled Elements, Tertiary Text)
  - **#bababa** (Borders & Subtle Dividers)
  - **#f7f7f7** (Card Backgrounds, Light Mode Sections)

These colors ensure **strong contrast**, **readability**, and **brand consistency** across different devices and screen sizes.

3. Typography & Iconography

3.1. Font Selection

- **Primary Font**: **Jost** (Modern, Clean, and Readable)

This font provides **aesthetic consistency** while maintaining **high readability across desktop and mobile interfaces**.

3.2. Icon Set

The website uses **Lucide Icons & Svg Icon** for a **modern and scalable** icon set. Icons are used for:

- **Navigation Menu & Categories**
- **Cart, Wishlist, and Checkout Steps**
- **Social Media & Interactive Features**
- **Product Status (In Stock, Out of Stock, Discounts, etc.)**

All icons are used in **SVG format** to ensure scalability and performance optimization.

3.3. UI Components & Design Principles

3.1. Navigation & Layout Structure

- **Fixed Header, Scroll to Top button & Tiker** to improve accessibility.
- **Top Navigation Bar (Top-Nav) Includes**:
  - Contact Information (Phone, Email).
  - Free Shipping Information.
- **Primary Navigation Includes**:
  - Products, About, Media, Support, Activities.
  - Search Bar, Cart, and Vistor counter Shortcuts.

3.2. UI Components & Interactions

- **Smooth Hover Effects** on buttons and menus.
- **Interactive Filters & Sorting** on product pages.
- **Collapsible Sections** in FAQ and Policies.
- **Animated Loading Elements** for better user engagement.

3.3. Mobile & Responsive Design

- **Fully Responsive Grid System** (Using **CSS Flexbox & Grid Layouts**).

- **Optimized Navigation Drawer** for mobile users.
- **Touch-friendly Buttons & UI Elements** for smooth user interactions.

3.4. Design Tools & Prototyping

4.1. Design & Prototyping Tool

The entire UI/UX design process was carried out in **Figma**, ensuring a streamlined workflow between **designers and developers**. The Figma file includes:

- **Wireframes & High-Fidelity Mockups**.
- **Interactive Prototypes for Testing**.
- **Component Libraries for Consistency**.
- **Style Guide & Design Tokens**.

4.2. Shared Figma Link

Figma Design File:

MediFlow project evidence: source repository, requirements, architecture, event, and progress documents.

(Accessible for developers, stakeholders, and UI/UX teams).

Important Note:

The Figma file serves as a reference only. During development, there may be **deviations** from the original design due to technical constraints, optimization decisions, or responsive adjustments. The final coded version **may not be 100% identical** to the Figma design.

4. Website Structure Overview: Header, Footer, and Layout Sections

Before diving into the detailed layout of each page, it's important to first define and understand the **Header** and **Footer** components, as they are crucial parts of the website's structure. These sections ensure **consistent branding, seamless navigation, and accessibility** across different pages.

- We strive to optimize the UI/UX as much as possible, ensuring a smooth, intuitive, and enjoyable user experience.

4.1. Header Section



The **Header** is the topmost part of the website and plays a critical role in user interaction. It serves as the **first touchpoint** for visitors, providing key navigation and interactive elements.

Top Navigation Bar (Top-Nav)

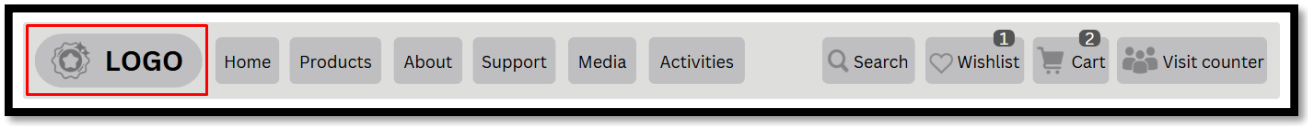


- **Positioned at the very top of the website, above the main navigation menu.**

- **Left Side:** Displays **company contact information**, including:
  - **Phone Number**
  - **Email Address**

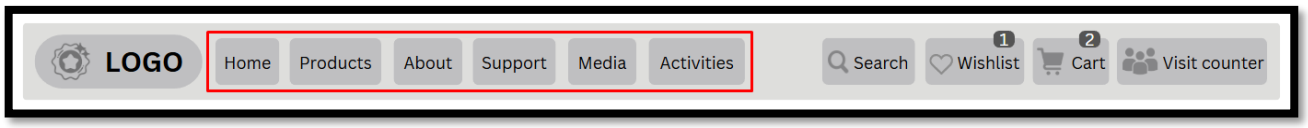
- **Right Side:** Displays **free delivery conditions**, indicating the **minimum purchase amount required** to qualify for free shipping.

Company Logo



- Positioned at the **top-left** of the page.
- Clickable and redirects users to the **Homepage** when clicked.

Main Navigation Menu & Development Plan



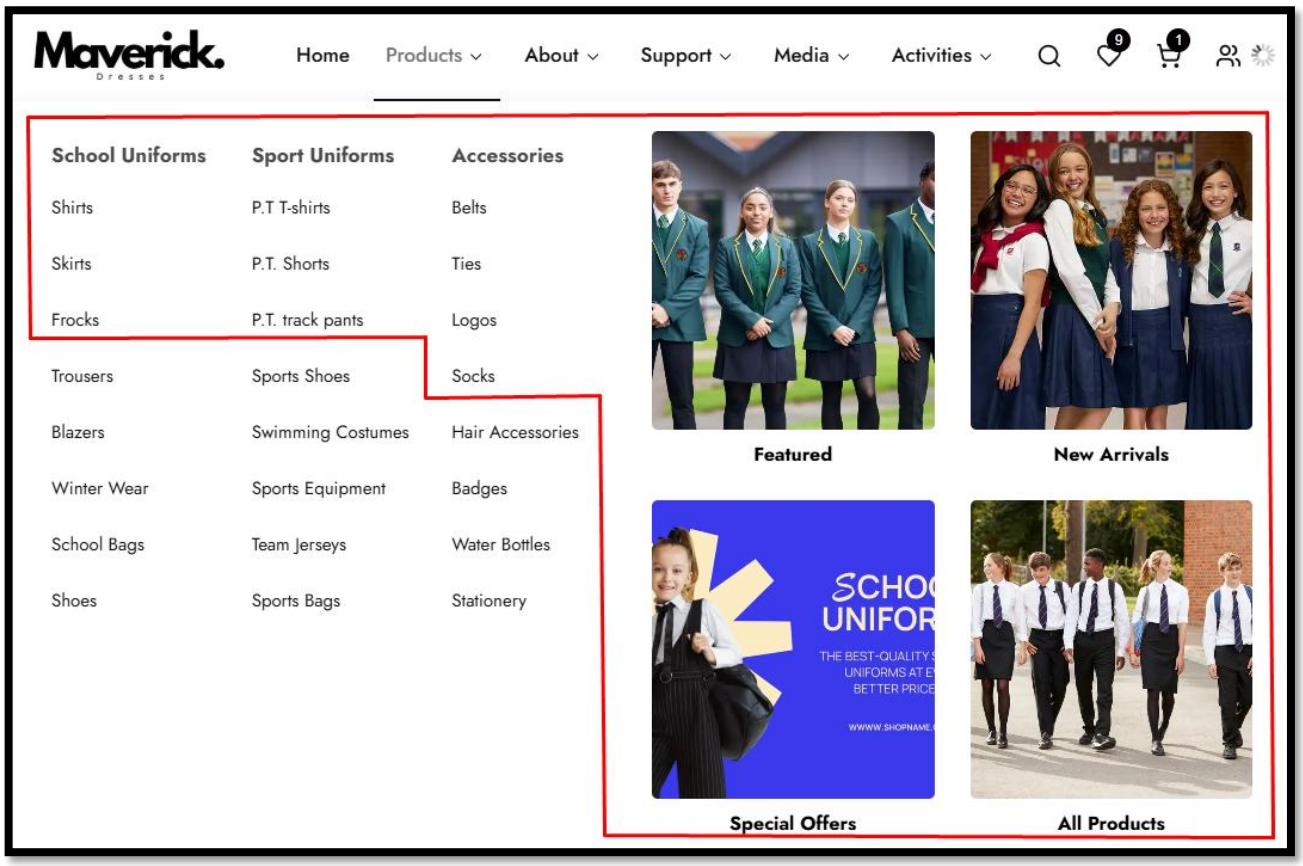
- The **Main Navigation Menu** is positioned at the **top-left** of the page and serves as the primary method for users to navigate through the Maverick Dresses website efficiently. It features a **horizontal** or **vertical menu bar** that provides quick access to major sections.

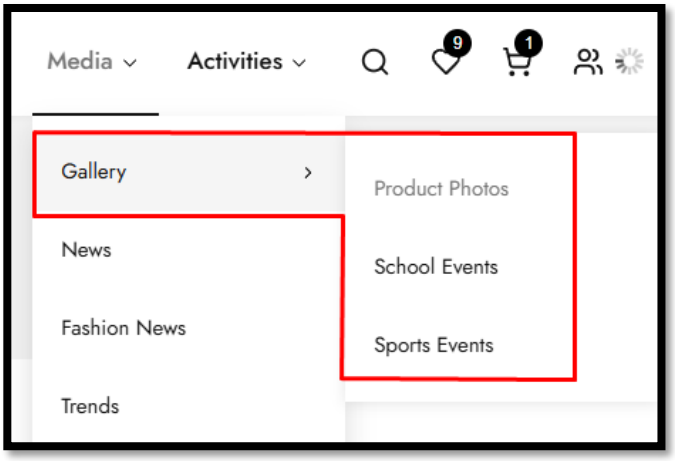
o A horizontal or vertical menu bar that allows users to quickly access major sections:

- **Products** (Featured, New, Sale, School Uniforms, Sports Uniforms, Accessories)
- **About** (About Us, Careers, Partners, Customer Growth Chart, Awards)
- **Media** (Fashion News, Gallery)
- **Support** (FAQ, Policies, Feedback, Contact Us)
- **Activities** (Wishlist, Coupons, Order Tracking)

- **Development Plan (Products & Media Navigation):** To ensure a structured and optimized development process, the Products and Media sections in the Main Navigation Menu are being implemented in two phases.

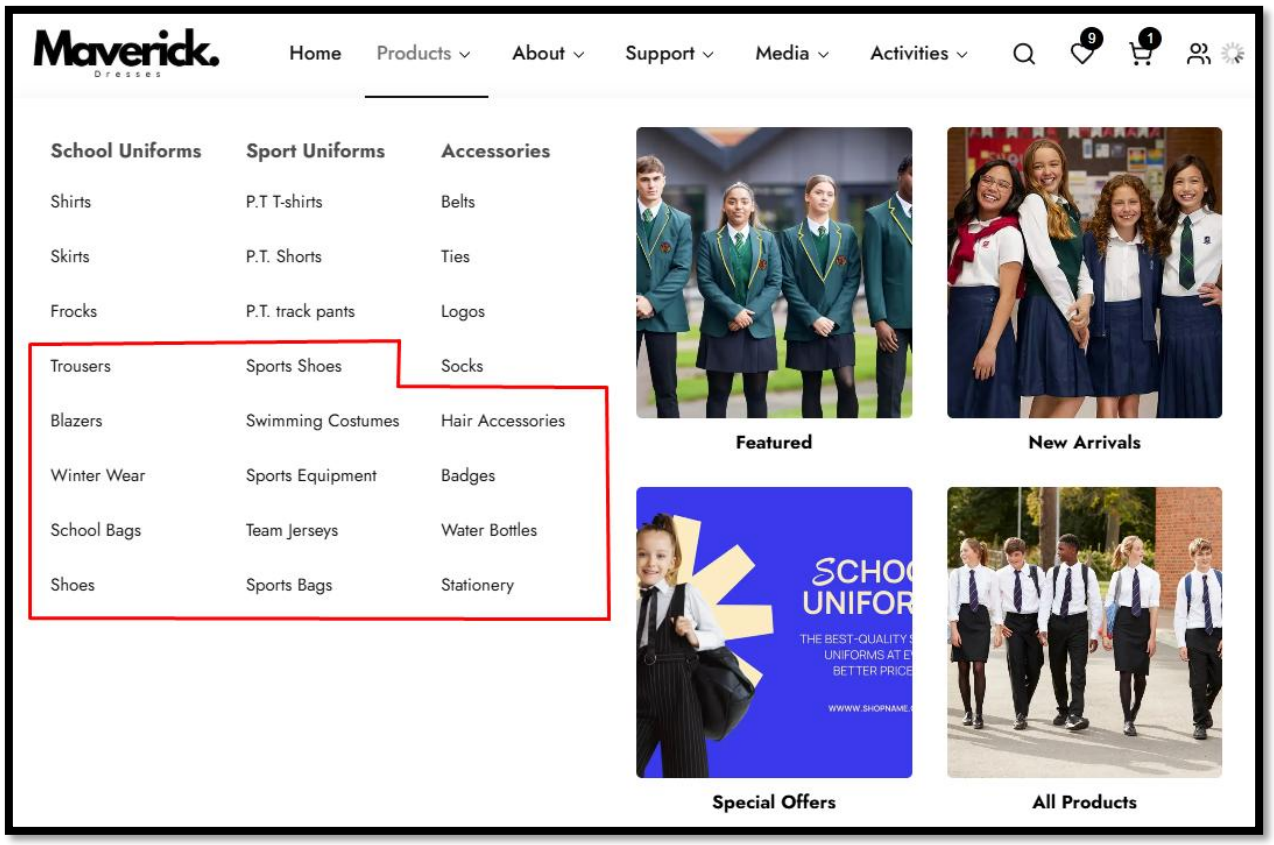
  - **Version 1 - Completed Features**

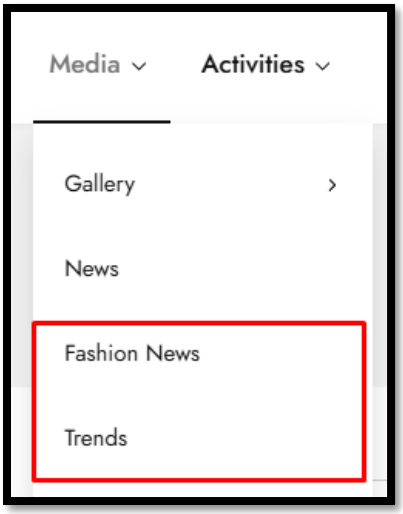




- In **Version 1**, the following **Products** categories and **Media** sections have been fully implemented, with all links functional:
- **Products** -> Featured, New Arrivals, Special Offers, All Products, School Uniforms (Shirts, Skirts, Frocks), Sports Uniforms (P.T. T-Shirts, P.T. Shorts, P.T. Track Pants), Accessories (Belts, Ties, Logos, Socks).
- **Media** -> Gallery (Product Photos, School Events, Sports Events), News

- **Version 2 - Upcoming Enhancements (Currently Not Clickable)**





- The following **sections** are planned for **Version 2**, and their links are **not yet active**:

- **Products** -> Advanced Filtering & Sorting Features for better user experience. Expanded Accessories Section with additional product lines. Detailed Subcategories for School Uniforms & Sports Uniforms, including: (More variations for Shirts, Skirts, and Frocks. Additional uniform styles for different school levels. More Sportswear categories such as custom jerseys, sweatpants, and training gear.).
- **Media** -> Fashion News and Trends – A dedicated section covering worldwide industry updates, style guides, and the latest collections from Maverick Dresses.

- **Version 2 - Upcoming Enhancements (Currently Not Clickable)**
- **By dividing development into two phases, Maverick Dresses ensures:**

- A functional and navigable website in Version 1.
- A scalable and enriched user experience in Version 2.
- A seamless transition between current and future updates.
- With **Version 1 successfully implemented**, the upcoming enhancements in **Version 2** will further elevate the website's **product navigation and media content**, making it a **fully immersive shopping and branding platform**.
- Stay tuned for Version 2!

- **Navbar Highlight for Active Page:** The **Navigation Bar (Navbar)** helps users easily navigate between different sections of the Maverick Dresses website. To improve **user experience and usability**, the active page should be **highlighted** dynamically, allowing users to see which section they are currently visiting.

- **Active Page Indication:**

- The currently selected **menu item** should have a **distinct color change** and/or an **underline** to indicate the active state.
- Example: If a user is on the **Products Page**, the "Products" menu item should be visually

highlighted.

- **Hover Effects:**

- When users **hover over menu items**, a **soft color change** or **underline animation** should appear to indicate interactivity.

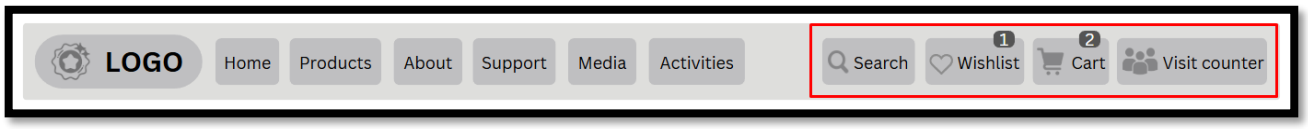
- **Dropdown Menu Active State:**

- If a user selects a **sub-menu item**, both the **main menu and the selected sub-menu item** should be highlighted.
- Example: If a user is on the **FAQ Page**, the "Support" menu (parent category) should also be highlighted.

- **Smooth Transitions:**

- Implement a **fade-in or slide-in animation** when users switch between pages to enhance UX.

Right Icons Menu



Search Bar Icon: Click to show Search bar.

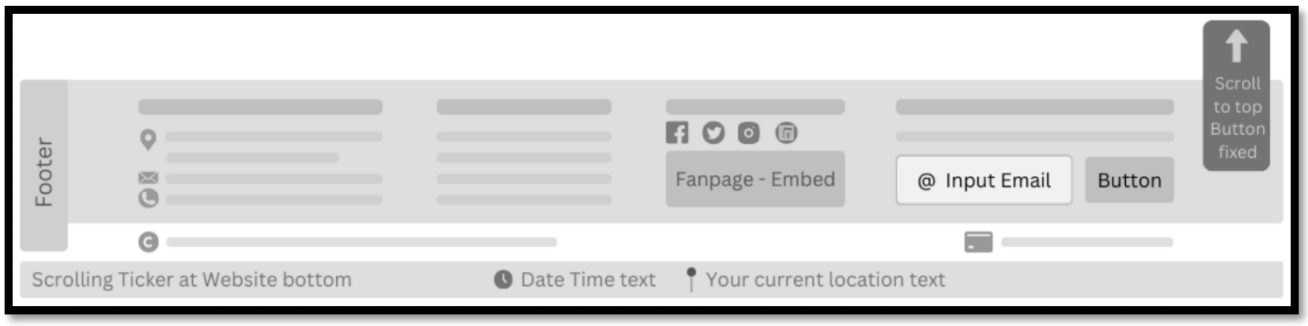
Wishlist Icon: Displays the number of items in the Wishlist.

Cart Icon: Displays the number of items in the cart.

Visitor Counter

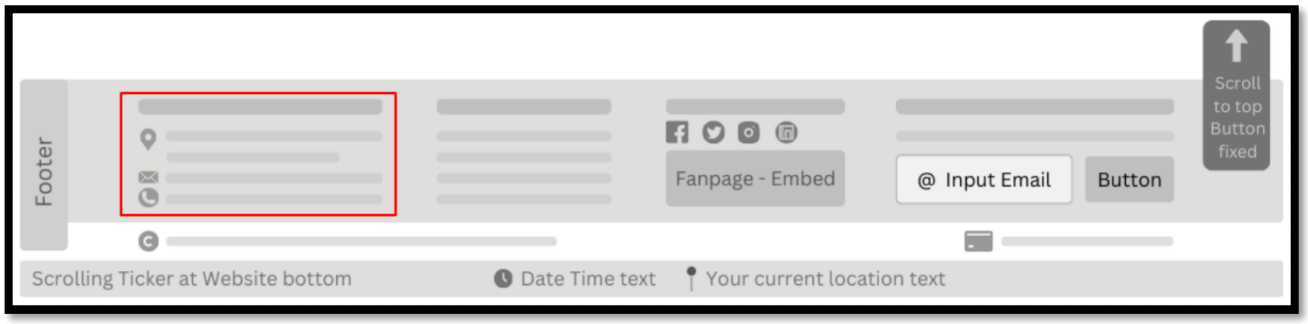
- Positioned near the **logo or within the header section**.
- A **real-time counter** displaying the number of active visitors on the site, creating a **sense of community and engagement**.

4.2. Footer Section



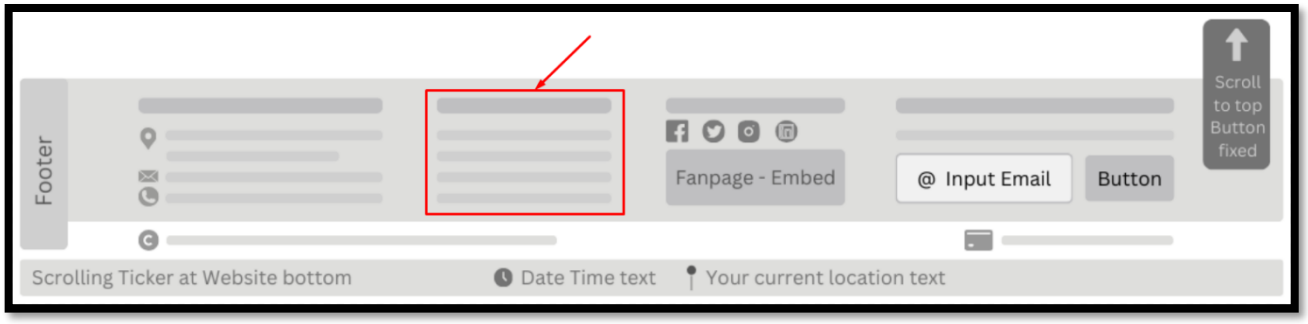
The **Footer** appears at the bottom of each page and provides additional resources for users, such as company details, policies, and quick navigation.

Company Information



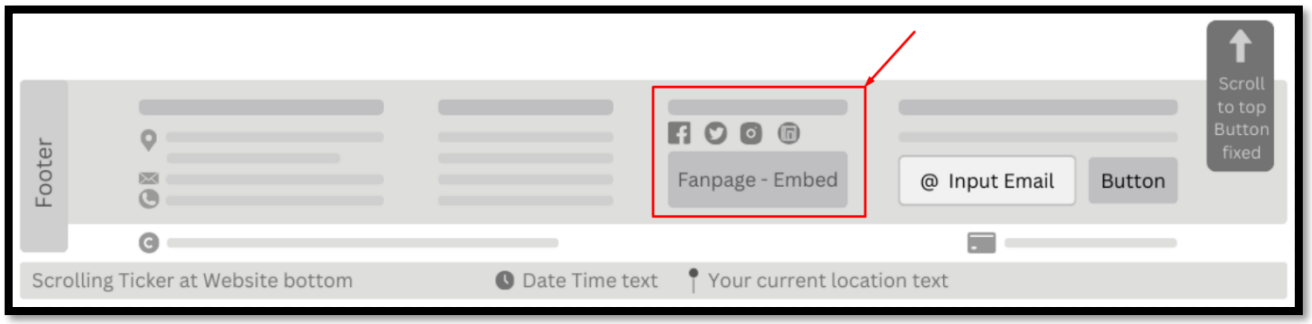
- Displays essential details such as company name, physical address, phone number, and email address.
- May also contain a **link to the About Us page** or any other relevant company information.

Quick Links



- Provides **direct access** to commonly accessed pages such as:
  - Policies
  - FAQ
  - Careers
  - Sitemap
  - Partners
  - Contact Us

Social Media Icons and Links

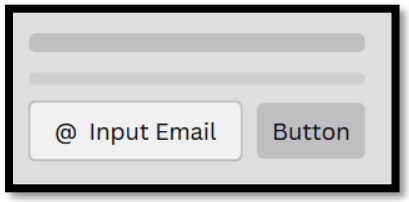


- Allows users to connect with the brand on various platforms, such as:
  - Facebook
  - Instagram

- Twitter
- LinkedIn

- The footer includes an embedded Facebook fanpage, allowing users to:
- View recent posts.
- Follow Maverick Dresses directly from the site.
- Engage with brand updates without leaving the page.

Newsletter Signup



- Includes an email input field where users can subscribe to newsletters and promotional offers.

Payment Methods



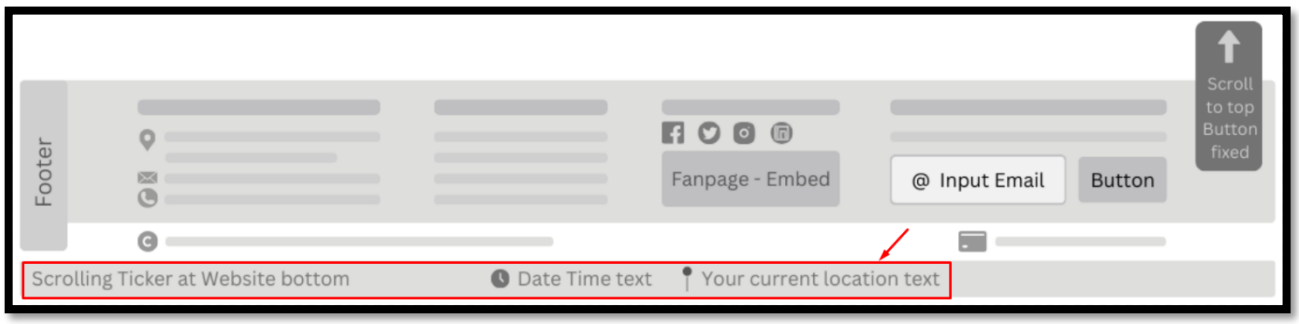
- Displays **supported payment options** for transparency, such as:
- Visa
- MasterCard
- PayPal
- …

Copyright Notice



- Located at the **very bottom of the footer**, displaying ownership rights and legal disclaimers.

Scrolling Ticker Section



- A **persistent ticker located at the very bottom of the page**.
- Make sure there is a button that asks the user to give permission for the browser to know the user's current location. In case the user forgets to give permission when first entering the website, the button will be displayed.
- Displays **real-time information**, including:
  - **Current Date & Time**
  - **User's Geolocation (City, Country)** (via **Geolocation API**).

Scroll to Top Button



- Positioned at the **bottom-right corner of the page**.

- A **fixed floating button** allowing users to instantly scroll back to the top of the page.
- Enhances **usability, especially for pages with long content**.

4.3. Page Layout Sections Overview

The following sections define the content structure within different pages, helping organize information effectively and improving navigation.

Hero Section (Banner)

- The **first section users see** on the homepage.
- Contains **large images or sliders** showcasing:
  - Promotions
  - Best-selling products
  - Limited-time discounts
- Includes call-to-action buttons such as *Shop Now* or *Discover More*.

Main Content Section

- Central area of each page containing relevant content.
  - Product pages display product images, descriptions, and filtering options.
  - The About Us page includes the brand's history, values, and mission.
  - Support pages provide company policies, FAQs, and customer assistance information.

Sidebar (Optional, for Some Pages)

- Typically present on **product listing pages or blogs**.
- Includes filtering options such as price, size, category, and availability.

- May also display related promotions or featured content.

Interactive Features Section

- Enhances user experience with elements such as dynamic sliders, accordions for FAQs, and live updating product widgets.

Call-to-Action (CTA) Section

- Encourages user interaction through newsletter signups, add-to-cart buttons, and limited-time offer banners.

4.4. Important Notes

Product Titles & Breadcrumbs:

- If a **product title** is too long, it will be **automatically truncated with "..."** after three lines to **prevent layout distortion**. To view the full title, users must **click on the product** for more details.
- **Breadcrumbs** will also be truncated if they exceed a certain length to **maintain a clean and structured navigation flow**.

4.5. Layout & Responsiveness

The website is **fully responsive**, ensuring **optimal performance across mobile devices, tablets, and all major web browsers** as per the project requirements. The design dynamically adjusts to different screen sizes, providing a **seamless and user-friendly experience** on **Chrome, Firefox, Safari, Edge, and other modern browsers**.

Next Steps

With the **Header, Footer, and Key Layout Sections** defined, the next step is to break down the structure of individual pages in more detail, starting with the **Homepage Layout**.

Testing document

1. Authentication and Access - User Input Validation

No.	Feature Tested	Remarks
1	**Username or email**	Required; trim whitespace; reject an unknown or disabled account.
2	**Password**	Required; invalid credentials return a safe error without revealing account state.
3	**Role and token**	JWT, expiry, and endpoint role are checked before protected access.
4	**Refresh session**	Refresh is accepted only for a valid session

		and is rejected after account disablement.

2. Organization and Account - User Input Validation

No.	Feature Tested	Remarks
1	**Department name**	Required; trim whitespace; reject duplicate names within the organization.
2	**Staff and role**	Required staff identifier and an allowed role; role changes are traceable.
3	**Account status**	Only valid active/inactive transitions are accepted.
4	**Search and pagination**	Filters are bounded and

		lists use the agreed default and maximum page size.

3. Patient - User Input Validation

No.	Feature Tested	Remarks
	Patient identifiers	Required fields are validated; duplicate patient identifiers are rejected.
	Contact and demographic data	Formats and permitted lengths are checked before save.
	Search result	Only permitted patient fields are returned to the

		requesting role.

4. Appointment - User Input Validation

No.	Feature Tested	Remarks
1	**Patient and clinician**	Both identifiers are required and must exist in the approved context.
2	**Date and time**	Required format; invalid or conflicting appointments are rejected.
3	**Status transition**	Only an approved lifecycle transition is accepted.
4	**Cancel/check-in**	Action is role checked and the resulting state is visible to relevant users.

5. Clinical Record and Diagnosis - User Input Validation

No.	Feature Tested	Remarks
	Clinical episode	Appointment and patient linkage is required before a record is saved.
	Required clinical fields	Validation covers the fields needed for the current record type.
	Diagnosis	Diagnosis is linked to the correct episode and

		authorized clinician.

6. Laboratory - User Input Validation

No.	Feature Tested	Remarks
1	**Laboratory request**	Test type, patient, episode, and requester are required.
2	**Processing status**	Only valid request-to-processing transitions are accepted.
3	**Result values**	Values and required conclusion fields are validated before completion.
4	**Result publication**	Author, time, permissions, and notification behavior are checked.

7. Pharmacy and Inventory - User Input Validation

No.	Feature Tested	Remarks
1	**Medicine and quantity**	Required; quantity must be positive and within available stock.
2	**Prescription eligibility**	Patient, prescription, payment, and status are checked before dispensing.
3	**Concurrent stock**	No negative stock and no duplicate issue under

		retry or concurrent requests.
4	**Failure event**	Insufficient stock and processing failure return a traceable outcome.

8. Billing, Payment and Dispensing - User Input Validation

No.	Feature Tested	Remarks
1	**Invoice**	Only approved

		charges and a valid open invoice can be paid.
2	**Payment amount**	Amount and method are required and validated.
3	**Saga correlation**	Payment and dispensing share a correlation ID and explicit states.
4	**Compensation**	Downstream failure produces compensation or a manual-review state.

9. Notification and Reporting - User Input Validation

No.	Feature Tested	Remarks

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1	**Recipient**	Notification is addressed only to the permitted owner.
2	**Duplicate event**	Repeated event does not create a duplicate business notification.
3	**Message retrieval**	User sees only messages allowed by role and ownership.
4	**Report filters**	Date and organization filters return explainable approved totals.

10. Inpatient and Surgery - User Input Validation

No.	Feature Tested	Remarks
	Admission or case	Patient and responsible clinician are required.
	Bed or schedule	Conflicts and invalid transitions are rejected.
	Role and ownership	Extension actions follow the same role and service ownership rules.
	Outcome trace	State, actor, timestamp, and correlation information remain available.

11. Web/Mobile Client - User Input Validation

No.	Feature Tested	Remarks
1	**Client session**	Unauthenticated access is redirected or rejected according to route policy.
2	**Loading and empty states**	The UI communicates progress and no-result conditions clearly.
3	**Error handling**	Validation and server errors are actionable without exposing

		sensitive details.

Final checklist

No.	Review 01 aspect tested	Suggestion / Remarks
1	Project scope covers organization, patients, appointments, clinical records, laboratory, pharmacy, billing, notifications, and reporting.	Review 01 evidence prepared / To be checked
2	The team roster, supervisor, class, and submission date are confirmed in the project records.	Review 01 evidence prepared / To be checked
3	The problem statement identifies duplicated data, unclear status, manual handover, stock risk, saga risk, and limited visibility.	Review 01 evidence prepared / To be checked
4	Actors and roles are documented: ADMIN, MANAGER, DOCTOR, NURSE, LAB_TECH, PHARMACIST, CASHIER, PATIENT, and SYSTEM.	Review 01 evidence prepared /

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		To be checked
5	The solution identifies the API Gateway, service discovery, bounded services, PostgreSQL ownership, and RabbitMQ messaging.	Review 01 evidence prepared / To be checked
6	Authentication and authorization requirements are written with invalid and unauthorized cases.	Review 01 evidence prepared / To be checked
7	Patient identity and duplicate prevention requirements are written.	Review 01 evidence prepared / To be checked
8	Appointment lifecycle and invalid transition requirements are written.	Review 01 evidence prepared / To be checked

9	Clinical record, diagnosis, laboratory, and result traceability requirements are written.	Review 01 evidence prepared / To be checked
10	Pharmacy stock, concurrency, idempotency, and dispensing requirements are written.	Review 01 evidence prepared / To be checked
11	Billing, payment, dispensing, saga, and compensation requirements are written.	Review 01 evidence prepared / To be checked
12	Notification ownership and duplicate-event behavior are written.	Review 01 evidence prepared / To be checked
13	Reporting dimensions and explainable totals are written.	Review 01 evidence prepared /

		To be checked
14	Hardware and software requirements identify Java 21, Maven, PostgreSQL, RabbitMQ, Next.js, React, TypeScript, and pnpm.	Review 01 evidence prepared / To be checked
15	Non-functional baselines cover security, data ownership, performance, resilience, consistency, observability, testing, and usability.	Review 01 evidence prepared / To be checked
16	The Review 01 boundary excludes insurance, PACS/DICOM, laboratory devices, production bank integration, and multi-region recovery.	Review 01 evidence prepared / To be checked
17	Repository alignment is recorded for backend contexts, gateway, discovery, Web pages, tests, Mobile readiness, and documentation.	Review 01 evidence prepared / To be checked

18	Review 01 validation tables are completed for the core user inputs and failure paths.	Review 01 evidence prepared / To be checked
19	The team records remaining approval items for production SLA, RTO/RPO, log retention, and accessibility target.	Review 01 evidence prepared / To be checked

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Tasksheet

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No	Member	Contents	Page Name	Activity Plan	Date of Preparation of Activity Plan		
				Planed Start Date	Actual Start Date	Actual Days	Status
1	All members; Vinh consolidates	Review outstanding functions, agree API contracts and integration order. Produce a shared list of changes, permissions and test data for web and mobile.	Backend / Frontend / Mobile — scope and API alignment	2026-10-01	2026-10-04	5	**In progress** Design and APIs exist; web/mobile integration alignment remains.
2	Hoàng Anh	Complete Core account, staff, department and patient management. Enforce login and record permissions; keep Organization and Patient data independent.	Backend — Core: Organization, Patient	2026-10-01	2026-10-04	10	**Awaiting integration** Core and APIs exist; complete login and cross-service data flows remain unverified.

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3	Hoàng Anh	Complete Gateway routing and caller identity checks. Expose authorized APIs and reject forbidden requests or expired sessions.	Backend — Gateway: authentication, routing, authorization	2026-10-02	2026-10-06	9	**In progress** Routing and role checks exist; payment-request discovery routing and prescription cancellation permissions remain open.
4	Vinh	Complete appointment, check-in, examination start and completion workflows. Enforce valid state transitions and required payment clearance.	Backend — Clinical: appointments and medical records	2026-10-01	2026-10-05	12	**Awaiting integration** Business logic and tests exist; acceptance with actual examination payments remains open.

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5	Vinh	Complete lab orders, sample reception and result delivery. Link orders to the correct record, synchronize results and prevent duplicates on retry.	Backend — Lab: orders, samples and results	2026-10-02	2026-10-06	12	**Awaiting integration** Orders, results and recovery tests exist; integrated lab payment acceptance remains open.
6	Huy	Complete drug, inventory, prescription and dispensing workflows. Check stock and payment; enforce cancellation and dispensing permissions and prevent duplicate dispensing.	Backend — Pharmacy: drugs, stock and prescriptions	2026-10-01	2026-10-04	12	**Awaiting integration** Drug and dispensing workflows exist; Gateway role checks and the paid prescription journey remain open.

7	Lộc	Complete charges, invoices and payment recording. Collect the correct examination, lab and prescription charges; ensure accurate amounts and prevent duplicate payments.	Backend — Billing: charges, invoices and payments	2026-10-01	2026-10-05	12	**In progress** Charges, payments and tests exist; event publication evidence and consumer acceptance remain incomplete.
8	Huy	Complete reports and verify data sources. Provide supported filters and reconcilable figures; identify unavailable data clearly.	Backend — Report: reporting	2026-10-04	2026-10-07	10	**Awaiting integration** Report APIs exist; source coverage and data completeness await confirmation.

9	Lộc	Complete in-app notifications and recipient access checks. Deliver to the correct recipient, protect patient information and prevent duplicate delivery on retry.	Backend — Notification: notifications	2026-10-04	2026-10-08	10	**Awaiting integration** Notification APIs exist; actual event delivery and recipient permissions await integrated acceptance.
10	Huy, Lộc; Vinh coordinates; Hoàng Anh supports Gateway	Complete Support packaging and service connections. Start Core, Clinical, Lab, Pharmacy, Billing and Support with correct routing and data connections.	Backend — Support and integration configuration	2026-10-05	2026-10-09	10	**Not started** No Support host module exists; Report and Notification remain separate services.

11	Each business owner	Test backend workflows, permissions and failure recovery. Cover successful requests, forbidden access, invalid states, duplicates, connection failures and retries; fix discovered defects.	Backend — business workflow testing	2026-10-07	2026-10-11	11	**In progress** Business and recovery tests exist; system-wide integration testing remains open.
12	Vinh coordinates; Hoàng Anh supports accounts	Review login, role-based menus, layout and small-screen behavior. Handle session expiry and ensure usable navigation on phones.	Frontend — login, navigation and shared layout	2026-10-01	2026-10-05	7	**In progress** Layout, navigation and login exist; real-session and role-based navigation checks remain open.

13	Hoàng Anh	Complete staff, department and patient screens. Support authorized search, pagination, viewing and updates, with input validation and API error handling.	Frontend — /organization; /patients	2026-10-03	2026-10-06	11	**Awaiting integration** Management pages and actions exist; verification against Core through Gateway remains open.
14	Vinh	Complete appointment, check-in and medical-record screens. Run the examination workflow through Gateway and show correct states and payment conditions.	Frontend — /appointments; /records	2026-10-03	2026-10-07	11	**Awaiting integration** Appointment and record screens exist; payment-dependent examination actions await acceptance.

15	Vinh	Complete lab-order and result screens. Display orders and update states and results for the correct record and authorized roles.	Frontend — /lab; /records	2026-10-04	2026-10-07	11	**Awaiting integration** Screens and local tests exist; the paid lab flow through Gateway remains unverified.
16	Huy	Complete drug, stock and dispensing screens. Search drugs, view stock and process prescriptions according to backend states and server-provided prices.	Frontend — /pharmacy	2026-10-03	2026-10-07	11	**Awaiting integration** Drug, stock and prescription screens exist; real-role and paid dispensing checks remain open.

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17	Lộc	Complete invoice and payment screens. Select the correct charges, view invoices and record payments; prevent repeated submissions and show clear errors.	Frontend — /billing	2026-10-03	2026-10-06	12	**Awaiting integration** Invoice and payment screens exist; source-based charge discovery through Gateway remains unavailable.
18	Huy	Complete report screens. Match filters and figures to the backend and handle loading, empty and error states.	Frontend — /reports	2026-10-05	2026-10-08	10	**Awaiting integration** Report screens and local tests exist; actual figures and source coverage remain to be reconciled.

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19	Lộc	Complete notification screens. Show lists and details to the correct recipient and expose only backend-supported actions.	Frontend — /notifications	2026-10-05	2026-10-09	10	**Awaiting integration** List, detail and send screens exist; recipient checks with real sessions remain open.
20	Each business owner	Test screens by role and business workflow. Use real APIs, verify permissions and error handling, and record defects.	Frontend — role-based screen testing	2026-10-07	2026-10-11	11	**In progress** Local screen tests exist; complete role coverage and real-API journeys remain unverified.

21	Vinh coordinates; all members review workflows	Review the existing mobile app, agree screens and roles, and configure connections. Confirm the runtime environment and use Gateway for all requests.	Mobile — app configuration and screen scope	2026-10-01	2026-10-04	6	**In progress** Folder scaffolding and a loading component exist; app configuration and Gateway connectivity are missing.

22	Vinh coordinates shared work; Hoàng Anh supports authentication	Complete login, secure session storage and role-based navigation. Handle logout and session expiry and restrict access to authorized screens.	Mobile — login, logout and navigation	2026-10-02	2026-10-06	8	**Not started** Mobile login, session storage and navigation have not been implemented.
23	Hoàng Anh	Complete patient search and information screens. Use authorized APIs and roles and protect personal information.	Mobile — patient information	2026-10-04	2026-10-07	10	**Not started** Mobile patient screens and data handling have not been implemented.

24	Vinh	Complete appointment and examination information screens. Show authorized schedules, states and records; use only supported update APIs.	Mobile — appointments and examination information	2026-10-04	2026-10-08	10	**Not started** Mobile appointment and examination screens have not been implemented.
25	Vinh	Complete lab-result screens. Link results to the correct patient and record and handle refresh, empty and error states.	Mobile — lab results	2026-10-05	2026-10-08	10	**Not started** Mobile lab-result screens have not been implemented.
26	Huy	Complete drug and prescription information screens. Display medicines, instructions and prescription states according to backend data and user permissions.	Mobile — drugs and prescriptions	2026-10-04	2026-10-08	10	**Not started** Mobile drug and prescription screens have not been implemented.

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27	Lộc	Complete invoice and payment-status screens. Show correct charges, amounts and states; expose payment actions only when supported by the backend.	Mobile — invoices and payment status	2026-10-04	2026-10-07	11	**Not started** Mobile invoice and payment screens have not been implemented.
28	Lộc	Complete user notification screens. Display notifications for the correct recipient and prevent access to another account’s notifications.	Mobile — notifications	2026-10-05	2026-10-09	10	**Not started** Mobile notification screens have not been implemented.

29	Each business owner tests; Vinh packages the app	Test mobile on devices and prepare an Android installation package. Check screen sizes, keyboards, slow or lost connections and session expiry; verify installation and startup.	Mobile — device testing and Android package	2026-10-07	2026-10-11	11	**Not started** No complete app, device testing or Android package is available for acceptance.
30	All members; Hoàng Anh prepares accounts	Prepare role-based accounts, sample data and shared test scenarios. Use consistent data across backend, web and mobile without real personal information.	Backend / Frontend / Mobile — accounts and test data	2026-10-06	2026-10-09	10	**In progress** Backend/web sample data and test fixtures exist; a shared set covering mobile is not ready.

31	All members; Vinh coordinates	Test the complete check-in, examination, lab, payment and dispensing journey. Run through Gateway on web and agreed mobile screens and reconcile states, amounts and data.	Backend / Frontend / Mobile — examination-to-dispensing workflow	2026-10-08	2026-10-12	11	**Awaiting integration** Actual payment journeys depend on pending Gateway work; mobile workflows are not implemented.
32	All business owners	Verify reports, notifications, permissions and consistency across applications. Reconcile figures and states, protect data and prevent duplicate effects when events are retried.	Backend / Frontend / Mobile — reports, notifications and permissions	2026-10-08	2026-10-12	12	**Awaiting integration** Cross-application data, permissions and events have not passed integrated acceptance.

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33	Each business owner	Fix integration defects and retest affected workflows. Resolve login, examination, payment, dispensing and security blockers and record remaining limitations.	Backend / Frontend / Mobile — defect fixes and regression testing	2026-10-09	2026-10-13	11	**In progress** Local fixes and regression tests exist; integration issues and dependencies remain to be checked.
34	All members ; Vinh consolidates	Complete installation and user guides, project reports and presentation materials. Document working functions, configuration, demo accounts and presentation scenarios.	Documentation — installation, usage, reports and presentation	2026-10-09	2026-10-12	12	**In progress** Design documents and guides exist; handover materials need the final delivery results.

35	All members	Package, perform final acceptance and hand over the project. Rerun main scenarios, verify the installation package and environment, and deliver source, configuration, documentation and remaining limitations.	Backend / Frontend / Mobile — acceptance and handover	2026-10-10	2026-10-14	11	**Not accepted** System-wide acceptance and the complete handover package are not yet available.

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Acknowledgments and Declaration

Project Documentation for Mediflow

Completed on 30/10/2025

The documentation for the **Mediflow** project was completed on November 30, 2026. Our team would like to express our gratitude to our supervisor, **Mr. HOÀNG ĐỨC QUANG**, for his guidance and valuable feedback throughout the project. We also appreciate the constructive input from our fellow students, which contributed to the refinement of our work.

We also thank **Project Aptech** for providing us with the opportunity to work on this real-world project, which helped us apply our academic knowledge in a practical scenario.

This document is the result of our team's efforts, based on the knowledge we have gained during our studies and additional research. We confirm that the content is original and accurate, with no unauthorized copying or plagiarism. Our team is committed to maintaining academic integrity and ensuring the quality of this project.

This is a complete document that reflects the efforts and research of our team as students who are still learning and have not yet entered the professional workforce. We are grateful to our teacher for his support throughout this project. The content is based on the knowledge we have acquired during our studies and reliable sources.